S1CogBench: What System One Decision Models Can and Cannot Do on Cognitive Experiments

Dengzhe Hou[1,*] **Kazunori D Yamada**[1]

[1]Tohoku University

[*]Corresponding author. Project: github.com/dengzhe-hou/S1CogBench

Abstract

Language models are no longer used only to generate text. Software also asks them for decisions, in routing, guardrails, and automated gates. System One decision models, also called typed decision models, such as Jev and Laya, return one fast decision over the legal options, leaving no text to inspect. However, a fast, well-formed decision does not show that the model used the evidence. Cognitive experiments can test this, because they vary task demands and measure how behavior changes, yet existing cognitive benchmarks evaluate language and multimodal models. We introduce S1CogBench to ask what these models can and cannot do on 24 cognitive tasks, evaluating five open models and Jev and keeping conditions and scoring apart from disclosed adaptations. We find that decision models vary widely in reliability, acting well does not imply predicting humans, aggregate scores hide how decisions respond to conditions, and speed comes mainly from answering without generation. To our knowledge, S1CogBench is the first benchmark of System One decision models on established cognitive experiments, and it gives a shared test of whether their decisions respond to evidence, feedback, memory load, and future opportunities before software relies on them.

1 Introduction

Jev, Laya, and Intern-Decision exemplify a shift from free text generation to structured decisions (TypeSafe, n.d.; Convai Innovations, n.d.; Zhu et al., 2026). They return one typed decision over the legal options in a single call, so software can act on it directly for routing, guardrails, and automated gates (Zhuang et al., 2026; Rafe and Das, 2026), and a small model of this kind can be invoked many times inside a larger system. We call them *System One decision models* (TypeSafe, n.d.), also called typed decision models (Sun et al., 2026b; Sharma et al., 2026), as an operational label. We take their speed as a premise and measure where it comes from.

A typed interface makes a response easy to execute, but it does not show that the response uses the relevant evidence, and no generated text is left to inspect (Sharma et al., 2026). A model can return a well-formed distribution yet ignore feedback or stop searching too early. In closed-loop environments small System One models rarely succeed (Chen et al., 2026), but such environments do not isolate which demand fails. Cognitive experiments can, because they manipulate information, uncertainty, memory load, incentives, or future opportunities and show how behavior changes.

Existing cognitive benchmarks evaluate language and multimodal models (Coda-Forno et al., 2024; Binz et al., 2025; Ying et al., 2026; Nie et al., 2025; Hou et al., 2026b), and CogGym's 50 models include no System One decision model. System1-Arena does evaluate System One decision models, but on static decision tasks and agent environments (Chen et al., 2026). To our knowledge, S1CogBench is the first benchmark to evaluate System One decision models on established cognitive experiments.

We ask what these models can and cannot do on these experiments. RQ1 asks whether a model that acts well also predicts humans. RQ2 asks what overall scores hide about how behavior changes across conditions. RQ3 asks where the speed comes from.

Four findings answer these questions.

- Decision models vary widely in reliability. Laya does not beat uniform choice on 17 of 26 targets, against 3 for Intern-Decision, which also beats the Qwen3.5 backbone it is fine-tuned from on 23, the one pair we can compare (Section 4.2).
- Acting well does not imply predicting humans (RQ1). In Mouselab, Qwen has the highest autonomous reward but predicts humans worse than

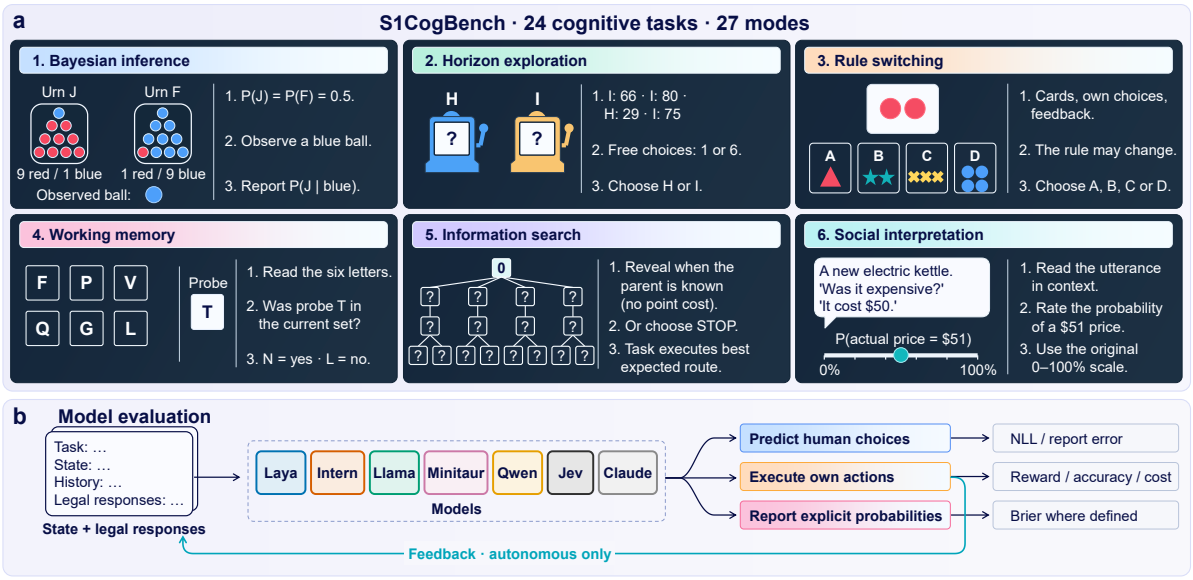


Figure 1: **S1CogBench connects cognitive tasks to model evaluation.** **(a)** Six representative tasks from the 24-task, 27-mode suite. **(b)** Adapters supply visible states, histories, and legal responses to the models, and each output has its own score. Model-specific coverage and API configurations are recorded in Appendix A.

uniform choice, and Laya has the best NLL of the open models with zero reward (Section 4.3).

- Aggregate scores hide condition effects (RQ2), as Horizon, the Wisconsin Card Sorting Test (WCST), and reinforcement learning and working memory (RLWM) show (Section 4.4).
- Speed comes mainly from the interface (RQ3). Under the same readout, Intern-Decision is no faster than its same-size backbone, while generation takes 80 to 316 times as long as one forward pass in the two general models we timed (Section 4.5).

S1CogBench is built for models that return a fast decision in a single call. It turns each of 24 tasks into a typed request over legal options, scores option probabilities and native choices, times each decision with generation as a control, and supports new model adapters.

2 Related work

Cognitive benchmarks evaluate language and multimodal models. CogBench uses ten behavioral metrics from seven experiments (Coda-Forno et al., 2024), CogGym evaluates 50 models on 258 single-turn experiments (Ying et al., 2026), and Psych-101 and Centaur evaluate held-out human-response prediction and feed model-generated choices back into Horizon and Two-step (Binz et al., 2025). CogArena studies the consistency and predictive validity of cognitive measurements (Hou et al., 2026b), and WMF-AM probes cumulative state tracking (Hou et al., 2026a). Namazova et al. (2025) compare predictive and generative behavior, and S1CogBench extends this distinction to explicit event probabilities, source-defined condition effects, and structured interfaces.

System1-Arena compares System One decision models through typed probability requests, closed-loop environments, latency, and cost (Chen et al., 2026), and TypedBench examines calibration, wording and option sensitivity, length, and asymmetric decision costs (Sharma et al., 2026). These works already provide unified interfaces, conditional diagnostics, and efficiency measurement. Table 1 compares the closest resources with ours, and Appendix D covers further work.

3 The S1CogBench benchmark

S1CogBench maps an original cognitive experiment to the information visible at each decision, its legal responses, a model output, a readout, and source-grounded scores. Figure 1 summarizes this connection.

3.1 Task selection and provenance

The 24 tasks fall into ten organizational domains (Table 2). WCST, Horizon, and Mouselab each have a human-prediction and an autonomous mode, giving 27 modes. Appendix B lists all 24 tasks and their sources. Each specification identifies the original study, adopted public dataset or environment version, manipulation, stimuli, legal responses, feedback, reset and stopping rules, and

Resource	Cognitive experiments	Predicts human responses	Closed-loop actions	System One models	Decision time or cost
CogBench (Coda-Forno et al., 2024)	7	–	✓	–	–
CogGym, Sept. 2026 v1 (Ying et al., 2026)	258	✓	–	–	–
Psych-101 / Centaur (Binz et al., 2025)	160	✓	(✓)	–	–
CogArena (Hou et al., 2026b)	13	–	–	–	–
BanditBench / EVOLvE (Nie et al., 2025)	–	–	✓	–	–
System1-Arena (Chen et al., 2026)	–	–	✓	✓	✓
TypedBench (Sharma et al., 2026)	–	–	–	✓	✓
DecisionPivot (Zhuang et al., 2026)	–	–	–	✓	–
S1CogBench	**24**	✓	✓	✓	✓

Table 1: Coverage of the closest evaluation resources in each paper's main evaluation. Counting units differ across resources, and (✓) marks selected tasks only. This compares coverage, not scores.

scoring unit. The 24 tasks come from published experiments and benchmarks, such as Horizon (Wilson et al., 2014), WCST (Grant and Berg, 1948), RLWM (Collins and Frank, 2012), Mouselab (Callaway et al., 2022), and the two-step task (Daw et al., 2011). Official splits are used where provided. Inclusion follows complementary manipulations and interface compatibility, independently of observed model rankings.

3.2 Typed decision interface

For each decision, an adapter gives the model what the original task shows at that point, the history h_t as a visible state x_t, and the ordered legal responses $\mathcal{A}_t$. For human prediction the history is the participant's, and for autonomous execution it is the model's own. Histories can be complete, summarized, windowed, or absent, and these are different information conditions. Responses can be categorical actions, ordinal ratings, numerical reports, or event probabilities.

The model returns a distribution $p_\theta(a \mid x_t, \mathcal{A}_t)$ over those responses. A readout is the rule that turns this output into a probability or an executed action, such as a native choice, a sampled action, or generative decoding. We record the distribution, the native choice, the executed action, the feedback, and the timing, so a model's distribution stays separate from randomness added at execution. An episode ends as in the original task, for example after a fixed number of trials, an explicit stop, or a terminal state.

Text displays, supplied histories, and constrained response sets are disclosed adaptations of the original tasks. Hybrid sampling supplies the authors' posterior summary, autonomous Mouselab has zero observation cost and picks the final route by expected value, and stationary bandits use the summary-history context of the original code, which exposes counts and mean rewards rather than the raw history.

3.3 Scoring targets

For observed participant i and scored trial t, human-response NLL scores the observed choice given its preceding history, using each source's response set and its participant, fold, or response weighting,

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i,t} \log p_\theta(a_{it} \mid h_{it}). \tag{1}$$

NLL over the legal responses and NLL over the full vocabulary are different scores. A returned zero gives infinite NLL unless the primary metric of the source scorer defines an epsilon, as in WCST and Horizon.

Autonomous scores use executed trajectories, with total reward

$$G = \sum_t (r_t - c_t), \tag{2}$$

where the reward, each cost, and the horizon follow the source environment. Single-event Brier $(q - Y)^2$ needs an explicit probability and an observed event. Answer-category Brier sums over categories, so the two are not on the same scale.

We report nine shared measures, accuracy, reward, pseudo-regret, MAE, human NLL, Brier, latency P50 and P95, and invalid-response rate, each where the task defines it. Per-mode definitions and tables, complete curves, per-target API records, and reference details are in the project repository. Original cognitive effects are reported alongside these scores. Uncertainty uses participants or independent environments, not trajectory steps, and paired 95% intervals state their resampling unit.

Domain	Tasks	What the experiments vary
Evidence	Bayesian urn, multiple-cue judgment, multi-attribute choice, probability coherence	Prior probability, likelihood, cue validity, coherence
Categories	SHJ, Medin, weather prediction	Category structure, feedback, transfer and typicality
History	RLWM, recent probes	Set size, repetition interval, recent interference
Adaptation	WCST, restless bandit, instrumental learning	Rule changes, changing rewards, outcome feedback
Exploration	Horizon, stationary bandits	Future opportunities, uncertainty
Information	Hybrid sampling, Mouselab	Reveal actions, sampling cost, stopping conditions
Preference	BART, temporal discounting	Risk, reward magnitude, explicit delays
Planning	Two-step	Transitions, changes in outcome value
Social	False-belief QA, politeness, pragmatic interpretation, nonliteral number	Belief access, communicative goals, literal meaning
Metacognition	Self-assessment	Difficulty, correctness

Table 2: The 24 tasks in ten organizational domains. Domains organize the tasks and are not validated latent factors. Appendix B gives each task's source.

3.4 References

Source-defined uniform controls are available for 26 performance targets, with varying legal supports and the original tie credit. References are fits to participants on 16 targets, other participants' responses on 5, policies without human fitting on 4, and analytic solutions or answer keys on 2, and Appendix A gives each one's fitting exposure and implementation.

4 Experiments

4.1 Setup

We evaluate five open models as released. Laya English, a non-autoregressive model built on ModernBERT, and Intern-Decision-4B are System One decision models. Intern-Decision is fine-tuned from Qwen3.5-4B (Qwen Team, n.d.) for structured decisions, which we call decision training, so Qwen is its backbone comparison. Llama-3.1-8B-Instruct (Grattafiori et al., 2024) is a general-purpose control, and Minitaur-8B (Binz, n.d.) is trained on human behavior. Their training data and compute are not matched. All 120 task–model pairs run on the full sample, and only multiple-cue judgment uses a compact history.

Jev, available only through its API, runs all 27 modes, and Claude5.5, a general model with adaptive thinking, runs selected tasks because of cost (Appendix A).

To separate models from readouts, we also run all five models on Horizon and WCST under both legal argmax and sampling.

4.2 Main results

Table 3 gives one primary target for each of the 27 task modes. Laya is not better than uniform choice on 17 of 26 targets, followed by Llama on 11, Minitaur on 7, Qwen on 6, and Intern-Decision on 3. Intern-Decision has the best score of the five models on 16 targets and beats its Qwen3.5 backbone on 23. The remaining mode, temporal discounting, reports how often the later reward is chosen, a preference with no better direction.

4.3 Does acting well imply predicting humans? (RQ1)

Figure 2 plots acting against predicting humans for the three tasks that have both modes. Mouselab separates the two most clearly. In this search task a model reveals tree nodes or stops, and the task then takes the path with the best expected value. We set the observation cost to zero, whereas the human data come from the original waiting-time protocol, so the two modes also differ in cost. Intern, Laya, and Minitaur stop immediately in all 23,750 episodes, so their reward is zero. Qwen reveals 15.5 nodes per episode and reaches 17.40, above the 11.77 of MetaGreedy, a one-step value-of-information policy replayed on the same trees (Appendix C). These are terminal expected values, not realized path rewards. Human prediction gives a different answer. Qwen's NLL is 2.464, all five models are worse than the 1.634 of uniform choice, and Laya is best at 1.717 with zero reward.

Horizon and WCST separate the two less clearly, and only under the original readouts. Among the five open models in Horizon (Table 4), the two scores mostly agree, although Minitaur has the lowest NLL (0.409) and Intern-Decision the highest reward (54.93). Under one shared readout, Minitaur also has the highest reward. A three-parameter Rescorla-Wagner (RW) fit to 25 training participants scores NLL 0.400 and reward 54.56.

Task	Uniform	Open models					API models		Reference
		Laya	Intern	Llama	Minitaur	Qwen	Jev	Haiku	
Human-response NLL ↓									
Multiple-cue judgment	2.197	2.342	**1.570**	2.687	1.984	3.012	$+\infty$	–	2.029
Multi-attribute choice	0.693	0.730	**0.546**	0.786	0.722	0.613	0.329	0.206	0.192
WCST	1.386	1.462	1.378	1.431	**1.344**	1.968	1.543	–	0.840
Horizon	0.693	0.702	0.518	0.503	**0.409**	0.661	0.414	0.390	0.400
Hybrid sampling	1.386	1.390	1.116	1.515	**1.039**	2.648	$+\infty$	–	0.264
Mouselab	1.634	**1.717**	2.109	2.743	2.125	2.464	1.458	–	1.367
Politeness	1.014	1.192	**0.799**	1.584	1.008	1.000	$+\infty$	$+\infty$	0.587
Pragmatic interpretation	1.282	1.592	**0.814**	1.851	1.049	1.240	$+\infty$	$+\infty$	0.416
Self-assessment	2.039	$+\infty$	2.029	2.523	**1.885**	2.262	$+\infty$	–	1.619
Accuracy under given histories (%) ↑									
SHJ categories	50.0	50.0	65.0	52.9	54.5	**71.3**	87.4	91.2	93.8
Medin categories	50.0	49.6	**72.3**	53.5	55.9	53.8	86.8	97.4	96.6
Weather prediction	50.0	54.5	**73.0**	47.5	53.8	50.5	80.8	80.2	78.2
RLWM	33.3	34.9	**77.8**	41.7	49.0	59.3	89.5	89.4	89.3
Recent probes	50.0	49.7	**82.1**	55.6	49.5	53.5	99.0	100.0	100.0
False-belief QA	50.0	39.5	82.5	81.8	65.0	**82.8**	71.5	82.6	100.0
Autonomous choices, correct or optimal (%) ↑									
WCST	25.0	25.5	27.9	**31.4**	28.2	25.2	28.5	–	48.9
Restless bandit	50.0	55.4	**65.6**	57.1	49.0	50.5	63.9	–	68.8
Instrumental learning	75.0	73.9	81.4	79.2	**86.7**	80.0	86.7	–	94.9
Horizon	50.0	54.8	**78.7**	66.4	71.7	63.3	82.9	–	74.6
Stationary bandits	12.5	17.6	**37.4**	13.5	9.8	28.9	47.0	–	68.1
Autonomous reward ↑									
Mouselab	7.01	0.00	0.00	3.77	0.00	**17.40**	8.21	–	11.77
BART	0.84	0.00	**0.74**	0.28	0.00	0.00	1.76	6.27	10.49
Two-step	0.50	0.50	**0.52**	0.50	0.51	0.52	0.55	0.55	0.52
Report error (MAE) ↓									
Bayesian urn	0.349	0.291	**0.235**	0.240	0.294	0.246	0.068	0.001	0.000
Probability coherence	32.13	32.98	**18.95**	28.57	19.04	21.87	17.66	–	16.54
Nonliteral number	34.77	30.86	**16.87**	25.10	28.53	29.75	20.37	22.52	4.98
Preference, later reward chosen (%)									
Temporal discounting	50.0	0.0	16.7	0.0	80.0	90.0	50.0	77.8	41.5

Table 3: Primary targets of the 27 modes, grouped by what is scored. Arrows show the better direction. Bold marks the best of the five open models, gray marks scores not better than Uniform, a dash means no complete score, and $+\infty$ means zero probability on a human choice under a scorer with no epsilon. Reference is a task-specific baseline, such as a fit to human data or an analytic solution. Open models keep their original readouts (Appendix A), and Haiku's NLL uses stated probabilities. Appendix B gives reference types and the reason for each dash, and Appendix A gives Opus and Sonnet.

In WCST, Minitaur predicts best (NLL 1.344), whereas Llama acts best (31.4%) but predicts worse than uniform (1.431 against 1.386). A shared argmax puts Minitaur first at 34.3%.

Probability scores also select different models. Among the open models in Bayesian urn inference, Intern has the lowest posterior MAE over 400 generated queries (0.2345), whereas Qwen has the lowest event Brier error (0.2299).

Jev scores at or near the top in RLWM and autonomous Horizon. It reaches 89.5% in RLWM under human-block histories, level with the 89.3% of the same-participant fitted reference under argmax, and 82.9% optimal choices in Horizon, between that task's human-fitted reference at 74.6% under sampling and 84.9% under greedy replay. Its two-decimal probabilities, however, give zero mass to some human choices, so its NLL is infinite on five of its nine prediction targets (Appendix A). None of those five has an autonomous mode. Where both modes exist, Jev beats uniform on both in Horizon and Mouselab (NLL 0.414 and 1.458). In WCST its NLL (1.543) is worse than uniform and its accuracy (28.5%) is close to it.

4.4 What do aggregate scores hide? (RQ2)

Horizon shows a response to future opportunities that the pooled NLL misses. When more choices remain (horizon 6 against 1), humans choose the less-observed arm 10.00 percentage points more

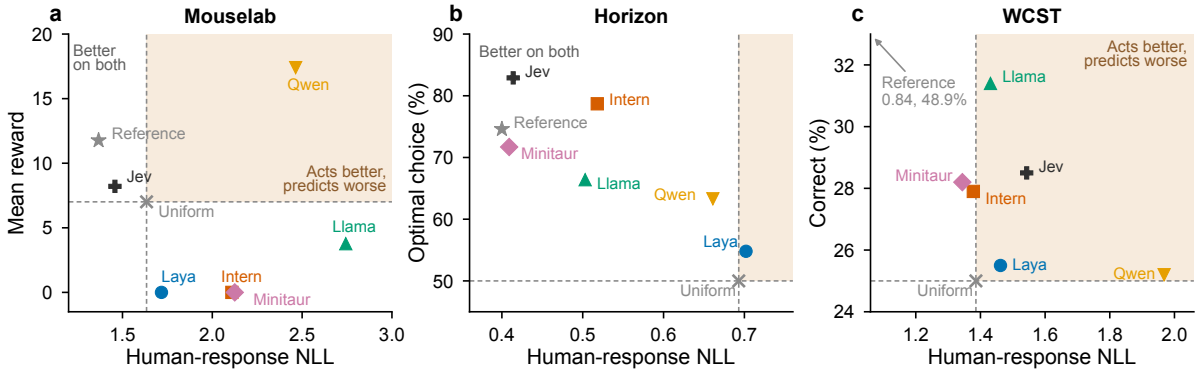


Figure 2: Human prediction against autonomous score under the original readouts, from Table 3. Dashed lines mark uniform choice, so the upper-left region beats it on both scores and the shaded region acts better but predicts worse. Panel c is zoomed, with Reference off scale. Points are not paired on the same decisions across the two axes.

often at their first unequal-information free choice. Under the same histories, models predict changes of only -0.49 to 1.18 points (Figure 3a).

In WCST, fewer perseveration errors can accompany worse accuracy. Laya makes them on 15.24% of trials against 28.82% for Llama, but its autonomous accuracy is also lower (25.47% against 31.39%), and its rate is close to the 17.13% of uniform choice, so this rate alone cannot rank flexibility (Table 5). All five autonomous accuracies lie between 25.20% and 31.39%, close to the 25% of uniform choice and far below the 50.00% of humans, and the sequential-learning (SL) reference reaches 48.92%. On 74 common decisions from two source participants, Claude models with adaptive thinking reach 63.51 to 66.22%, against 18.92 to 43.24% for the open models under shared legal readouts (Table 6). So the text conversion supports learning from feedback and does not alone explain the near-chance open-model scores.

In RLWM, average accuracy does not predict the load effect (Table 7). Qwen averages 59.39% and loses 20.34 points from set size 2 to 6, whereas Minitaur averages 49.09% and loses 11.34, and Intern averages 77.83% and loses 11.37, both close to the 12.74 of humans. Learning follows average accuracy more closely (Figure 4). From presentation 1 to 10 at set size 6, Intern gains 61.49 points, more than the 46.35 of humans, because it starts below them and ends above them. Laya stays flat, gaining only 0.21.

	Human	Mean reward			Optimal choice (%)		
Model	NLL	orig.	argmax	sample	orig.	argmax	sample
Uniform	0.693	50.0	50.0	50.0	50.0	50.0	50.0
Laya	0.702	51.0	51.8	51.0	54.8	57.9	54.8
Intern	0.518	**54.9**	54.9	52.2	**78.7**	78.7	62.8
Llama	0.503	52.9	53.6	52.7	66.4	70.9	65.2
Minitaur	**0.409**	53.9	**56.2**	**53.7**	71.7	**84.0**	**70.9**
Qwen	0.661	52.2	52.2	51.8	63.3	63.3	60.8
RW fit	0.400	54.6	56.0	54.6	74.6	84.9	74.6

Table 4: Horizon prediction of six held-out participants (NLL over legal responses, 6,720 free choices) and autonomous play on 31 source timelines (34,720 free choices per model and readout). Orig. is each model's own readout, argmax for Intern and Qwen and sampling for the others. Within a model, only the readout changes across columns. Bold marks the best open model in each column.

	Human history		Autonomous		
Model	Accuracy	NLL	Accuracy	Persev.	Set-loss
Human	50.00	–	50.00	7.32	1.88
Uniform	25.00	1.3863	25.00	17.13	17.13
Laya	25.53	1.4616	25.47	15.24	18.16
Intern	28.90	1.3781	27.90	21.25	15.84
Llama	**32.56**	1.4307	**31.39**	28.82	13.78
Minitaur	30.06	**1.3435**	28.16	21.85	14.61
Qwen	25.07	1.9677	25.20	15.88	17.88
SL refit	60.40	0.8402	48.92	9.47	4.63

Table 5: WCST means over 75 held-out participants (12,754 trials per mode). Accuracy and error rates are percentages of all trials. Persev. is a wrong choice that repeats the rule of a wrong previous choice, and Set-loss is a wrong choice that leaves the rule of a correct one. SL is a sequential-learning model fitted to the other 300 participants. NLL uses the source $-\log(p+10^{-12})$, and invalid outputs stay in the denominators. Bold marks the best of the five models in the first three columns.

In Bayesian urn inference, overall MAE hides a difference between conditions. All five models have higher error when the likelihood is the more informative input than when the prior is (0.28 to

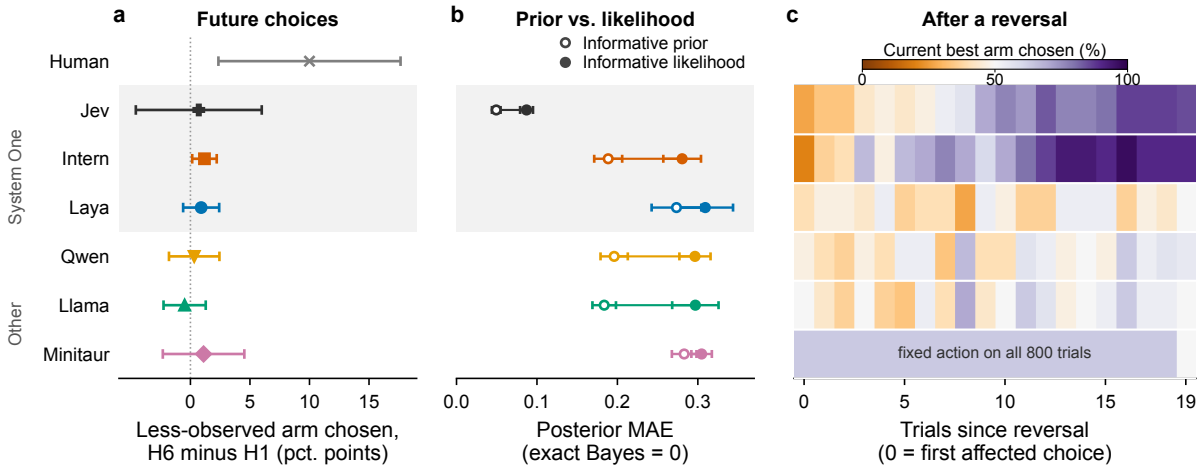


Figure 3: Cognitive profiles. (a) Predicted change in choosing the less-observed arm between horizons 6 and 1 under human histories. (b) Posterior MAE with a more informative prior or likelihood. (c) Choices aligned to reversals, as means over 10 runs. Minitaur selected arm F on all 800 trials, so its flat row is not adaptation. Error bars are 95% Student-t intervals over six participants (a) and 200 paired seeds (b), uncorrected for multiple comparisons. Panels b and c have no human data.

0.31 against 0.18 to 0.28, Figure 3b).

In the Restless bandit, a flat curve can reflect a fixed action rather than adaptation. Minitaur chooses arm F on all 800 trials, although its probability of F varies from 0.5622 to 0.9325, so its curve in Figure 3c reflects a constant action and the environment. Jev and Intern-Decision instead change after a reversal. They choose the current best arm on 36.7% and 42.0% of the first five trials after a reversal and on 85.0% and 91.3% of trials 15 to 19, whereas Laya, Llama, and Qwen reach 46.0%, 54.7%, and 55.0%.

	Horizon reward		WCST correct (%)	
Model	argmax	sample	argmax	sample
Opus	56.87		63.51	
Sonnet	56.95		66.22	
Haiku	56.86		66.22	
Laya	50.23	50.05	20.27	22.97
Intern	56.22	51.35	33.78	29.73
Llama	54.33	51.57	37.84	39.19
Minitaur	56.48	52.92	43.24	27.03
Qwen	52.68	51.33	18.92	22.97

Table 6: Saved decisions on common source identities, 200 Horizon free choices from one timeline and 74 WCST decisions from two participants. Claude answers in a final line of text with adaptive thinking, so the readouts are not matched, and Haiku's two invalid WCST attempts count as wrong. Each row keeps its own action-dependent history.

4.5 Where does the speed come from? (RQ3)

Under the same readout, the one size-matched pair shows no speed advantage from decision training. The controlled latency measurements cover nine configurations and 54 task-mode cells, with 37,056 timed requests on one A100 at batch size one (Table 8). With one forward pass over the legal options, latency grows with input length in every configuration. Intern-Decision, with the fused linear-attention (FLA) kernels used in the main evaluation, takes 0.75 to 1.41 times Llama's median end-to-end time on matched requests, close to its Qwen3.5 backbone at 0.71 to 1.40 times. Laya takes 0.67 to 0.95 times at FP32, 0.23 to 0.71 times at BF16, and 0.12 to 0.17 times on its TileLang fast path, but

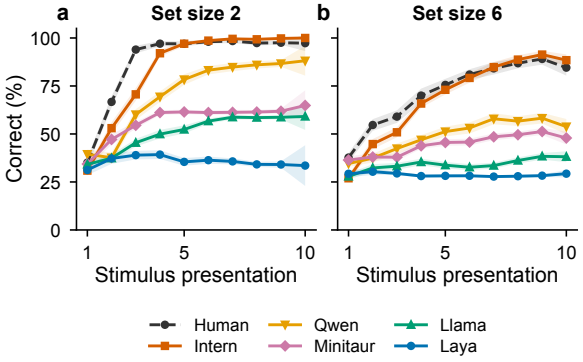


Figure 4: RLWM correct responses by stimulus presentation under complete human-block histories, with pointwise 95% Student-t intervals across participants (34 at set size 2 and 69 at set size 6 by presentation 10).

Source	Overall (%, $n = 79$)	Size 6 minus 2 (pp, $n = 79$)	Pres. 10 minus 1 size 6 (pp, $n = 69$)
Human	80.24	−12.74 [−14.54, −10.99]	+46.35 [41.13, 51.53]
Uniform	33.33	+0.00	+0.00
Laya	34.85	−7.17 [−8.53, −5.79]	+0.21 [−2.46, 2.69]
Intern	77.83	−11.37 [−12.80, −9.96]	+61.49 [58.09, 64.84]
Llama	41.78	−16.73 [−17.94, −15.46]	+10.49 [7.27, 13.62]
Minitaur	49.09	−11.34 [−12.58, −10.12]	+11.10 [7.34, 15.11]
Qwen	59.39	−20.34 [−22.21, −18.41]	+19.75 [15.71, 23.87]

Table 7: RLWM prediction of all 52,911 eligible responses from 79 participants under complete human-block histories. Accuracy uses the hidden correct action, and means weight participants equally, whereas Table 3 weights responses. Pres. means presentation, and contrasts pair the participants shown in the header. Brackets are pointwise 95% participant-bootstrap intervals (5,000 resamples, uncorrected).

with about 0.42 billion parameters it is far smaller than Llama.

The large gap lies between answering and generating. In a Horizon follow-up with a 4,096-token cap, median generation time is 2.958 seconds for Llama and 16.330 seconds for Qwen, against 36.97 and 51.66 milliseconds for their single forward passes. The extra time does not guarantee better decisions. In WCST, free-text generation under a 1,024-token cap brings Llama to 25.9% over all 12,754 actions, including 2,217 invalid responses, against 31.4% under its original readout, and Qwen reaches a 4,096-token cap on 75 of 130 attempts. The practical speed of a System One decision model therefore comes mainly from returning one typed decision without generation.

Through their APIs, Jev answers in a median of 0.35 seconds over all 27 modes, whereas Claude5.5 with adaptive thinking takes 2.60 to 3.59 seconds, at US$0.045 against US$0.34 to 8.96 per 1,000 decisions (Table 9). These client times include network and provider service time and cover different target sets, so they do not compare the models themselves.

Config.	Requests	E2E P50 (ms)	Ratio to Llama	Largest P95 (ms)
Laya FP32	5,576	25.50–274.44	0.67–0.95	742.54
Laya BF16	5,576	26.55–82.18	0.23–0.71	200.76
Laya fast	1,200	6.45–59.83	0.12–0.17	50,395
Intern FLA	5,576	52.06–219.40	0.75–1.41	437.76
Intern Torch	1,200	174.58–655.74	2.19–4.70	1,223
Qwen FLA	5,576	51.66–211.88	0.71–1.40	427.38
Qwen Torch	1,200	162.25–595.82	2.09–4.40	1,189
Llama	5,576	36.97–274.02	1.00	611.22
Minitaur	5,576	35.65–265.50	0.93–0.97	596.42

Table 8: Controlled latency over six task modes, on one A100 at batch size one. Ranges span the modes, and the ratio is the median end-to-end (E2E) ratio to Llama on matching requests. FLA uses the kernels of the main evaluation, and Torch runs without them. Laya fast is its TileLang path, whose 50,395 ms P95 on recent probes is consistent with compiling kernels during timed calls.

Model	Targets	Decisions	Latency (s) Median	Latency (s) P95	USD per 1,000	Output tokens
Jev	27	864,813	0.35	0.41	0.0448	94
Haiku	15	92,803	2.60	7.65	0.336	442
Sonnet	3	2,330	3.08	4.99	5.39	253
Opus	2	730	3.59	5.34	8.96	289

Table 9: API time and cost on each model's completed targets. Latency is the client round trip, so with different serving paths and targets it does not compare model speed. Output tokens count Jev's returned option probabilities and Claude's text with thinking.

4.6 Interface failures

Laya gives zero probability to 192 of 108,248 human responses in self-assessment, because its SDK rounds probabilities to four decimals, so its NLL there is infinite. Its true NLL is at least about 2.48, still worse than uniform. Minitaur reports 50% on 319 of 330 nonliteral-number items, a near-constant answer.

5 Conclusion

S1CogBench asks what System One decision models can and cannot do on 24 established cognitive tasks, because software can act on their decisions with no text to inspect. Decision models vary widely in reliability, acting well does not imply predicting humans, aggregate scores hide how decisions respond to conditions, and speed comes mainly from answering without generation. A fast, well-formed decision does not show that a model used the evidence, so these tests belong beside accuracy and speed before software relies on such models.

Limitations

The paper evaluates 24 tasks and three System One decision models, so the findings describe these tasks and models rather than the category, and the ten domains organize coverage rather than establish cognitive abilities. Text conversion, supplied histories, posterior summaries in Hybrid sampling, and zero-cost observation in autonomous Mouselab are disclosed adaptations. Readouts, training data, and compute differ across models, so the

analyses characterize checkpoints and interfaces rather than architectural causes. Public training descriptions cannot exclude overlap with cognitive datasets. The API comparisons are not matched in size, training, reasoning settings, or infrastructure, and Jev was not timed on matched hardware. Small human samples, such as six participants in Horizon prediction, widen participant-level uncertainty.

Ethical considerations

The study uses only published human data and environments under their original terms and recruits no new participants. Restricted participant records are not redistributed, and model-generated behavior is kept separate from human observations.

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A Protocol and reference details

Readouts. Horizon uses SDK probability sampling for Laya, the source sampled-token rule for Llama and Minitaur, and native legal argmax for Intern and Qwen. WCST uses legal argmax for Laya, Intern, and Qwen. Llama and Minitaur use full-vocabulary next-token argmax for human prediction and sampled generation for autonomous execution.

API models. Jev 1.14 is the `jev-1.14-20261014` snapshot. It executes its native choice, and its two-decimal probabilities are normalized without a floor. The Claude5.5 models use adaptive thinking, medium effort, and a 4,096-token cap and answer on a final line of text, and an invalid answer counts as wrong and ends a closed-loop run. Haiku's NLL uses probabilities it states in JSON, a different readout from the open models. Several stated politeness probabilities closely match the published human proportions, so exposure to these public data cannot be excluded. Opus and Sonnet score 0.001 on Bayesian urn and 15.34 and 20.81 on nonliteral number, and Sonnet scores 91.1% on Restless.

Reference	Fit on	How it is built
Exact Bayes	No human data	Analytic posterior for the generated urn trials
Horizon RW fit	25 training participants, six held out	Three-parameter formula fit, sampled autonomously with source seeds
WCST SL refit	300 training participants, 75 held out	Refit here, with separate prediction and autonomous histories
Mouselab MetaGreedy	95 participants, five folds	Official code, fit to human actions with 100 Sobol starts
Hybrid independent ANN	20 participants, 160 validation folds	3,000 training updates per fold
Multiple-cue update model	232 training and 28 test participants	Reimplemented from the source formula, updating only on visible feedback
Gaussian Thompson	No human data, 30 seeds in each of eight conditions	NIG prior and posterior, original environment and scoring
RLWM parameters	Fitted parameters of the source participants	Replayed from the archive, so not a held-out comparison with the models

Table 10: How selected references are built. The fitting data and the evaluated target differ across references, so they are not comparable training budgets.

Reference provenance. Analytic solutions, independent fits, author-code fits, disclosed reimplementations, and archived-parameter replays are distinct reference types. Table 10 summarizes references central to the reported findings.

B Notes on the main results

Table 3 scores. Each target comes from the full-suite modes or from a complete target of an API model. NLL keeps only the epsilons of primary source scorers, so WCST and Horizon stay finite where a zero probability would otherwise give $+\infty$. Minitaur's nonliteral reports use the number before its closing marker, because its generation continues past that marker. For multiple-cue judgment, Llama, Minitaur, and Jev use full prefixes and the other models use compact history.

Task mode	Task source	Reference type	Mark	Haiku
Multiple-cue judgment	Collsiöö et al., 2023	human fit	–	*n.a.*
Multi-attribute choice	Hilbig and Moshagen, 2014	human fit	†	–
WCST, prediction	Grant and Berg, 1948; Namazova et al., 2025	human fit	–	*n.a.*
Horizon, prediction	Wilson et al., 2014; Namazova et al., 2025	human fit	–	–
Hybrid sampling	D'Ambrogio et al., 2026	human fit	‡	*n.a.*
Mouselab, prediction	Callaway et al., 2022	human fit	‡	*n.a.*
Politeness	Yoon et al., 2016	human	¶	–
Pragmatic interpretation	Hu et al., 2023	human	¶	–
Self-assessment	Jansen et al., 2021	human fit	¶	*n.a.*
SHJ categories	Badham et al., 2017	human fit	†	–
Medin categories	Levering et al., 2020	human fit	†	–
Weather prediction	Speekenbrink et al., 2008	human fit	†	–
RLWM	Collins and Frank, 2012	human fit	§	–
Recent probes	Enkavi et al., 2019	human fit	†	–
False-belief QA	Le et al., 2019	optimal	–	–
WCST, autonomous	Grant and Berg, 1948; Namazova et al., 2025	human fit	–	*n.f.*
Restless bandit	Coda-Forno et al., 2024	algorithm	*	*part.*
Instrumental learning	Lefebvre et al., 2017	algorithm	¶	*part.*
Horizon, autonomous	Wilson et al., 2014; Namazova et al., 2025	human fit	–	*n.f.*
Stationary bandits	Nie et al., 2025	algorithm	†	*n.e.*
Mouselab, autonomous	Callaway et al., 2022	human fit	*	*n.e.*
BART	Frey et al., 2017	algorithm	¶	–
Two-step	Daw et al., 2011	human fit	¶	–
Bayesian urn	Coda-Forno et al., 2024	optimal	–	–
Probability coherence	Zhu et al., 2020	human	¶	*part.*
Nonliteral number	Kao et al., 2014; Tsvilodub et al., 2025	human	¶	–
Temporal discounting	Coda-Forno et al., 2024	human	¶	–

Table 11: Task sources, references in Table 3, and empty Haiku cells. Types are fits to participants (human fit), other participants' responses (human), analytic solutions or answer keys (optimal), and policies without human fitting (algorithm). Marks are an existing fit or policy (†), a same-participant fit (§), a within-participant holdout (‡), an adapted policy (*), and a value computed from source data (¶). Appendix C explains †, *, and ¶. Haiku cells are *n.e.* (not measured), *n.a.* (no full probability score), *n.f.* (partial prefixes only), or *part.* (incomplete records).

Uniform and references. Uniform averages over legal actions or reports with source weights and tie credit, using exact expectations for the autonomous modes rather than sampled runs. Temporal discounting uses 6 realized reward questions for Laya, Intern, and Llama and 10 for Minitaur and Qwen, without payment items, so its 50% uniform value holds only on those questions. False-belief QA has no human responses, so its reference is the answer key. Table 11 gives each task's source, each reference's type and source, and why a Haiku cell is empty.

C Sources of the main-results references

Mouselab and Restless. The references marked * adapt existing policies. Mouselab MetaGreedy reuses the 95 participant parameter sets archived for the policy family of Callaway et al. (2022), with no new fitting, and plays the same 23,750 synthetic trees as the five open models.

Restless runs a published two-parameter Q-learning model in the CogBench environment (Coda-Forno et al., 2024). Training reward on 100 independent environments selects $\alpha = 0.44$ and $\beta = 100$, the largest grid value, both fixed before test. Its source fits human responses instead, so this is not a human fit.

Computed from source data. The references marked ¶ are computed on CPU from the saved source data and original environments, with no model or API call, using the same targets, aggregation, and scorer as the five open models. Table 12 shows how each is built.

Existing fits and policies. The references marked † are existing fits or policies scored on the same items and targets. They are WADD for multi-attribute choice, GCM for SHJ, Medin, and weather prediction, the rational conditional-response model for recent probes, and the Thompson family for stationary bandits. WADD, GCM, and the rational model follow the published families (Binz et al.,

Target	Construction
Probability coherence	Median report of the 128 Psych-101 training participants for the same query.
Self-assessment	Lookup table of Binz et al. (2025) with add-α smoothing. The source has no baseline for the logic items, so these use tenfold cross-fitting across participants.
Politeness, pragmatic interpretation	Each human response is predicted from the other participants' responses to the same item with add-one smoothing.
Nonliteral number	MAE between the mean reports of two random halves of each item's respondents. Using half the respondents makes this reference conservative.
Temporal discounting	Mean later-reward rate of 13,232 Psych-101 participants on the same questions.
Instrumental learning	CogBench Rescorla-Wagner model with separate learning rates for positive and negative prediction errors (Coda-Forno et al., 2024; Lefebvre et al., 2017), selected on 100 independent training environments and fixed before test.
BART	One fixed pump threshold per balloon letter, selected on the same training environments. The thresholds stay below 14.00, the optimum when the hidden maxima are known.
Two-step	Seven-parameter hybrid model of Daw et al. (2011) fitted to each of the 40 CogBench human participants. Twenty-five of the 40 fits have at least one parameter at a bound, so they serve as a reference policy, not as estimates of individual learning.

Table 12: How the references computed from source data (¶) are built.

2025). The fits use source training participants and are reimplementations rather than the authors' own programs.

Recent probes shows the current letter set in the prompt, so its fitted model answers every item correctly, which does not show a working-memory mechanism. The stationary bandit conditions are the public FullBench conditions used for the open models, not the publication's horizon and seed protocol.

D Further related work

Cognitive benchmarks. CogToM builds a theory-of-mind benchmark from 46 paradigms (Tong et al., 2026), and EPITOME compares five language models with human baselines on six theory-of-mind experiments (Jones et al., 2024).

Predicting human behavior. Cross-Centaur predicts each participant's decisions in one of 12 psychological tasks from the same participant's behavior in another (Asano et al., 2026), and SocSci210 finetunes language models on 170 of 210 social science experiments to predict human response distributions (Kolluri et al., 2025). Sun et al. (2026a) find that context shifts LLM product choices in human-like ways but does not reliably reweight attention between features.

Sequential decisions. Horizon manipulates future opportunities and information asymmetry (Wilson et al., 2014), and BanditBench supports interactive choices with raw or summary histories and regret analysis (Nie et al., 2025). Mouselab and Tool-Lab study information acquisition and its consequences (Callaway et al., 2022; Wadi and Ma, 2026).

Other System One evaluations. Rafe and Das (2026) compare decision models with classifiers and general language models. DecisionPivot diagnoses probability ranking, calibration, and selective coverage with controlled task families (Zhuang et al., 2026), and DecisionRobust studies text and vision perturbations and prompt injection (Lan et al., 2026).